



# The purifier that knows its own cleaning power.

AdaptAir measures its real cleaning rate and filter condition in the room it's placed in, corrects its sensor for humidity, recognises whether pollution comes from an opened door or an indoor source — and keeps your air clean at the minimum measured energy. All on a single ESP32.

## AT A GLANCE

|            |                                            |
|------------|--------------------------------------------|
| Sensors    | PM · RH · CO <sub>2</sub> · door · power   |
| Filtration | Pre-filter + HEPA H13                      |
| Blower     | 12 V brushless DC, PWM                     |
| Display    | 0.96" OLED — PM2.5 · CO <sub>2</sub> · fan |
| Logging    | microSD (CSV) · Wi-Fi telemetry            |
| Compute    | ESP32 · int8 edge AI · no cloud            |

## MARKET POSITION

**\$579M**

INDIA MARKET '25

**13.5%**

ANNUAL GROWTH

**<5%**

HOMES OWN ONE

**₹10-15K**

SMART SEGMENT

**≥20%**

LESS ENERGY THAN AUTO-MODE

## WHY CHOOSE IT

### What makes AdaptAir different



#### Self-calibrating CADR

Fits PM2.5 decay after every clean-down and updates its real cleaning rate — live, in your room.



#### Real filter health

Fuses INA219 power signatures with measured CADR — separates a clogging filter from a leaky room.



#### Humidity-corrected sensing

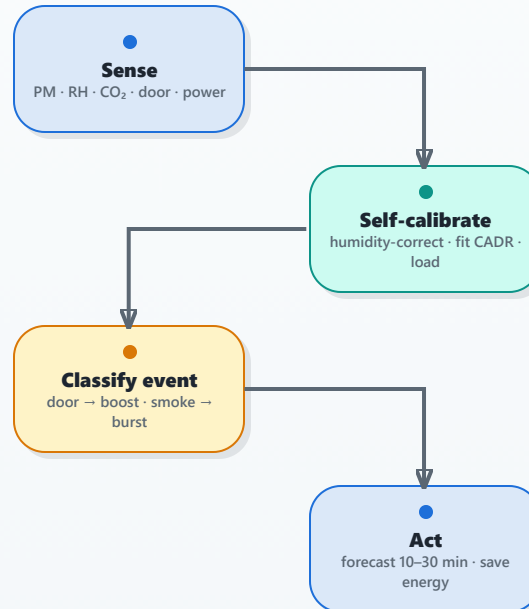
On-device BME280 RH correction gives trustworthy PM2.5 readings in humid monsoon and winter mornings.



#### Source-aware control

Reed-switch door events vs SCD40 CO<sub>2</sub> + PMS5003 size bins — incense gets a burst, a door gets a boost.

## CONTROL LOOP



## SMART AIR PURIFIER



# AdaptAir

Sense → self-calibrate → understand the event → act at minimum measured energy.

Self-calibrating

Source-aware

HEPA H13

Edge AI

IIITDM JABALPUR

DESIGN AND FABRICATION PROJECT – 2026